

SQL Statements to create tables and constraints:

```
CREATE TABLE business (  
    business_id CHAR(22) PRIMARY KEY,  
    name VARCHAR(60) NOT NULL,  
    address VARCHAR(75),  
    city VARCHAR(30) NOT NULL,  
    postal_code VARCHAR(7),  
    stars decimal(2, 1) CHECK (stars >= 1 AND stars <= 5),  
    review_count INT DEFAULT 0 CHECK (review_count >= 0)  
);  
  
CREATE TABLE checkin (  
    checkin_id INT PRIMARY KEY,  
    business_id CHAR(22) NOT NULL,  
    date DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,  
    FOREIGN KEY (business_id) REFERENCES business(business_id)  
);  
  
CREATE TABLE user_yelp (  
    user_id CHAR(22) PRIMARY KEY,  
    name VARCHAR(35) NOT NULL,  
    review_count INT DEFAULT 0 CHECK (review_count >= 0),  
    yelping_since DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,  
    useful INT DEFAULT 0 CHECK (useful >= 0),  
    funny INT DEFAULT 0 CHECK (funny >= 0),  
    cool INT DEFAULT 0 CHECK (cool >= 0),  
    fans INT DEFAULT 0 CHECK (fans >= 0),  
    average_stars DECIMAL(3, 2) CHECK (average_stars >= 1 AND average_stars <= 5)  
);  
  
CREATE TABLE tip (  
    tip_id INT PRIMARY KEY,  
    user_id CHAR(22) NOT NULL,  
    business_id CHAR(22) NOT NULL,  
    date DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,  
    compliment_count INT DEFAULT 0 CHECK (compliment_count >= 0),  
    FOREIGN KEY (business_id) REFERENCES business(business_id),  
    FOREIGN KEY (user_id) REFERENCES user_yelp(user_id),  
);  
  
CREATE TABLE review (  
    review_id CHAR(22) PRIMARY KEY,  
    user_id CHAR(22) NOT NULL,  
    business_id CHAR(22) NOT NULL,  
    stars INT NOT NULL CHECK (stars >= 1 AND stars <= 5),  
    useful INT DEFAULT 0 CHECK (useful >= 0),  
    funny INT DEFAULT 0 CHECK (funny >= 0),  
    cool INT DEFAULT 0 CHECK (cool >= 0),  
    date DATETIME DEFAULT CURRENT_TIMESTAMP,  
    FOREIGN KEY (business_id) REFERENCES business(business_id),  
    FOREIGN KEY (user_id) REFERENCES user_yelp(user_id),  
);  
  
CREATE TABLE friendship (  
    user_id CHAR(22),  
    friend CHAR(22),  
    PRIMARY KEY (user_id, friend),  
    FOREIGN KEY (user_id) REFERENCES user_yelp(user_id),  
    FOREIGN KEY (friend) REFERENCES user_yelp(user_id)  
);
```

Order of inserting data from CourSys into tables

```
USE rla122354
```

```
BULK INSERT business
FROM 'D:\userdata\rla122_assignment3\business.csv'
WITH (fieldterminator = ',', rowterminator = '\n', firstrow = 2)
```

```
BULK INSERT checkin
FROM 'D:\userdata\rla122_assignment3\checkin.csv'
WITH (fieldterminator = ',', rowterminator = '\n', firstrow = 2)
```

```
BULK INSERT user_yelp
FROM 'D:\userdata\rla122_assignment3\user_yelp.csv'
WITH (fieldterminator = ',', rowterminator = '\n', firstrow = 2)
```

```
BULK INSERT tip
FROM 'D:\userdata\rla122_assignment3\tip.csv'
WITH (fieldterminator = ',', rowterminator = '\n', firstrow = 2)
```

```
BULK INSERT review
FROM 'D:\userdata\rla122_assignment3\review.csv'
WITH (fieldterminator = ',', rowterminator = '\n', firstrow = 2)
```

```
BULK INSERT friendship
FROM 'D:\userdata\rla122_assignment3\friendship.csv'
WITH (fieldterminator = ',', rowterminator = '\n', firstrow = 2)
```